QP 25901341

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCIAA201 - SERVICES & SPECIFICATIONS I

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A Short Answer

Answer any 10 questions Each question carries 2 marks

1. **What is sound transmission loss?**
   Sound transmission loss is the reduction of sound passing through a material.
2. **What is wood wool?**
   Wood wool is a sound-absorbing material made from wood fibers.
3. **Differentiate between natural and artificial light.**
   Natural light comes from the sun; artificial light is produced by electrical sources.
4. **What is illuminance?**
   Illuminance is the amount of light falling on a surface.
5. **Define reflection of sound.**
   Reflection of sound is the bouncing back of sound waves from a surface.
6. **What is task-based lighting planning?**
   Task-based lighting planning provides lighting according to specific activities.
7. **Define circuit protection.**
   Circuit protection prevents damage from overloads and short circuits.
8. **What is glass wool?**
   Glass wool is an insulating material made from glass fibers.
9. **Suggest fire protection measures for a room with high fire load.**
   Use fire alarms, sprinklers, extinguishers, and fire-resistant materials.

10. What happens when an electrical circuit is overloaded?
Overloading causes overheating and may lead to fire or circuit damage.

11. Apply fire-retardant treatment to wooden furniture in interiors.
Apply fire-retardant coating or chemical treatment on wooden furniture.

12. Why is daylight important for sustainable design?
Daylight reduces energy use and improves comfort and health.

Part B Short Essay

Answer any 6 questions Each question carries 5 marks

13. Describe the diffusion of light and materials used for diffused lighting.
Diffusion of light is the even spreading of light. Frosted glass, acrylic sheets, and fabric shades are used for diffused lighting.

14. Write about casing, capping and batten wiring methods.
Casing wiring uses insulated wires inside wooden/PVC casing. Capping covers the casing. Batten wiring fixes wires on wooden battens using clips.

15. What are the factors that influence sound absorption?
Factors affecting sound absorption are material texture, thickness, density, and surface area.

16. Explain the safety regulations to be followed in residential electrical installations.
Use proper earthing, circuit breakers, insulated wires, correct load distribution, and avoid loose connections.

17. Explain the physical and psychological effects of good lighting in interiors.
Good lighting improves visibility, comfort, mood, productivity, and reduces eye strain.

18. Write an account about the building wiring system.
Building wiring system distributes electricity safely through wires, switches, sockets, DBs, and protective devices.

19. Apply the principles of fire protection while planning the interior layout of a small commercial office.
Provide emergency exits, fire extinguishers, alarms, smoke detectors, and fire-resistant materials in office interiors.

20. Explain the sound insulation of the floor and ceiling.
Floor and ceiling sound insulation is achieved using carpets, false ceilings, acoustic boards, and insulating materials.

Part C Essay

Answer any 2 questions Each question carries 10 marks

21. **Explain fire detection and alarm systems used in residential and commercial buildings. Discuss the working of smoke detectors and sprinkler systems.**

Fire detection systems include smoke detectors, heat detectors, and alarms. Smoke detectors sense smoke and activate alarms. Sprinkler systems release water automatically when heat rises above a set temperature. These systems help control fire and ensure safety in residential and commercial buildings.

22. **Explain major factors to be considered while designing lighting for various interior spaces such as classrooms, offices, and residences.**

Lighting design should consider room function, daylight availability, brightness level, glare control, energy efficiency, and user comfort. Classrooms need uniform lighting, offices require task lighting, and residences need both functional and decorative lighting.

23. **Explain various types of artificial lighting and their applications in interior spaces with suitable examples.**

Types of artificial lighting include ambient lighting, task lighting, accent lighting, and decorative lighting. Ambient lighting provides general illumination, task lighting is for specific work, accent lighting highlights objects, and decorative lighting enhances aesthetics. Examples include ceiling lights, table lamps, spotlights, and chandeliers.